

STUDENT ACTIVITY HANDOUT

IFY STATISTICS | SEMESTER 1 - WEEK 2A

Data and sampling



Name _____ Class _____ Date _____

Show working. Calculators allowed. Round percentages to two decimal places.

DISCUSSION

Planning the survey

Slide 6

D1. We want to know whether Eurasia students want more vegetarian dishes in the canteen. **What would make asking every student difficult?**

DISCUSSION NOTES

REFERENCE

Groups and their numbers

Slides 7-13

Term	Meaning
Population	The complete group we want to study.
Sample	The part of the population selected for the study.
Census	Collecting data from every member of the population.
Parameter	A numerical summary of the whole population, such as its mean or percentage.
Statistic	A numerical summary of a sample, such as its mean or percentage.

Start with the research question: which complete group do we want to describe?

CLASS NOTES

GROUPS AND THEIR NUMERICAL SUMMARIES

EXERCISE A1

Parameter or statistic?

Slides 14-18

Calculate each percentage. Classify it as a **parameter** or **statistic** and explain which group it describes. Round percentages to **two decimal places**.

- A1.1** A university club has **148 members**. All reply; **53** prefer Friday meetings. The research concerns **this club only**.

PERCENTAGE, TYPE AND REASON

- A1.2** A factory has **1,825 items**. It randomly tests **87**; **11 fail**. The research concerns **the whole batch**.

PERCENTAGE, TYPE AND REASON

- A1.3** Now use the **same club result** from A1.1 to describe the **whole university**. Does its classification change? Why?

- A1.4** Return to the factory: **1,825 items, 87 tested, 11 fail**. The remaining items have not been inspected.

(a) Smallest possible number of faulty items in the whole batch?

(b) Largest possible number? Explain what must be true of the uninspected items.

CALCULATION AND REASON

CALCULATION AND REASON

TYPES OF DATA

EXERCISE A2

Classify the variables

Slides 26-29

Use **C** = **categorical**, **QD** = **quantitative-discrete**, or **QC** = **quantitative-continuous**. Add a short reason.

Task / variable and example	Type	Short reason
A2.1 Programme code 1 = Business; 2 = Engineering		
A2.2 Rejected products in a batch 17 products		
A2.3 Journey duration, rounded for display Recorded as 43 minutes		
A2.4 Room number B-317		
A2.5 Customer calls received in a day 137 calls		
A2.6 Mass of food left over 83.6 g		
A2.7 Satisfaction category coded 1 to 5 1 = very dissatisfied; 5 = very satisfied		
A2.8 Exact age 18.73 years		
A2.9 Complete years of age counted at a date 18 complete years		
A2.10 Faults found in one cable 23 faults		
A2.11 Laboratory temperature 21.6°C		
A2.12 Telephone number Digits used to identify a phone		

Explain why the numerical appearance may mislead:

A2.1 Programme code:

A2.3 Rounded journey duration:

RANDOM SELECTION AND SYSTEMATIC SAMPLING

REFERENCE

Simple random sampling

Slides 31-32

Every possible sample of the chosen size has the **same chance**. Use a complete list, give each person a unique ID, and let a valid random generator choose **distinct IDs**. Match those IDs back to people.

EXERCISE A3

Systematic selection

Slides 39-41

A register has **2,184 students**. Select **84** systematically. Treat IDs as positions 1 to 2,184 in the register.

A3.1 Calculate the interval. State every possible random starting position.

INTERVAL AND VALID RANDOM STARTS

A3.2 A random generator gives **start 19**. Write the first six IDs, then find the **final selected ID**. Show your method.

FIRST SIX IDS AND FINAL-ID CALCULATION

A3.3 Calculate the sampling fraction as a simplified fraction, a decimal and a percentage.

FRACTION, DECIMAL AND PERCENTAGE

CHECK THE SELECTION PROCESS

EXERCISE A3

Reverse the process

Slides 39-41

A3.4 **Alternative plan:** use the same register of **2,184 students**. Select every **39th** student; the **fifth selected ID is 184**. Find the sample size and the starting ID. Explain.

SAMPLE SIZE, STARTING ID AND REASONING

EXERCISE A3

The favourite-student start

Slides 42-43

A3.5 The teacher uses the correct interval of **26** and says: **“My favourite student is at position 19, so I will start at 19 and keep adding 26.”**

What is wrong with this process? Why does the correct interval not repair it? Describe a valid way to choose the start.

PROBLEM, REASON AND REPAIR

DISCUSSION

Can the list cause problems?

Slides 44-45

D3. The register alternates Business (B) and Engineering (E). Start at **3** and select every **4th** position.

B	E	B	E	B	E	B	E	B	E	B	E	B	E	B	E	B	E	B	E
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

Which group does this sample miss? Explain why.

PUT IT TOGETHER - TRAVEL TO COLLEGE

EXERCISE A4

Travel to college

Slides 48-52

A fictional college has **2,460 enrolled students**. A valid random procedure selects **164**; all respond. **73** travel mainly by bus.

A4.1 Identify the population and sample. Calculate the percentage of respondents who mainly use the bus. Is it a parameter or statistic? Explain.

POPULATION, SAMPLE, PERCENTAGE, TYPE AND REASON

A4.2 Classify the three recorded variables.

Variable	Example	Type of data
Main transport mode	Bus	
Number of journeys per week	13	
Exact journey time	37.6 minutes	

A4.3 For a systematic sample of **164 from 2,460**, calculate the sampling fraction and interval. With **start 11**, write the first five IDs and the last selected ID.

SAMPLING FRACTION, INTERVAL, FIRST FIVE IDS AND FINAL ID

PUT IT TOGETHER - INTERPRETING ESTIMATES

EXERCISE A4

Estimates from samples

Slides 52-54

Continue with the same college: **2,460 students**; the first random sample contains **73 bus users out of 164**.

A4.4 Estimate how many students in the whole college mainly use the bus. Explain why your answer is not an established exact count.

CALCULATION AND INTERPRETATION

A4.5 A **second valid random sample** from the same college contains **86 bus users out of 164**. All selected students respond.

(a) Use the second sample to estimate the total number of bus users.

SECOND ESTIMATE

(b) Compare with your first estimate in A4.4. Must one calculation be wrong? Can we tell which estimate is closer to the true count? Explain.

EXPLANATION

PAIR CHECK

Check the reasoning

Slide 50

Check the two percentage denominators, the number of jumps to the final ID, and whether the bus totals are clearly labelled as **estimates**.

INDIVIDUAL EXIT CHECK AND OPTIONAL CHALLENGE

EXERCISE A5

Check your understanding

Slides 55-56

Answer independently. Give a reason as well as a number.

A5.1 We ask all **31 students in one class** about that class. Is this a census? What if our target is **the whole college**?

A5.2 A survey codes transport: **1 = bus, 2 = bike, 3 = walk**. What type of variable is this? Explain.

A5.3 Select **87 people from 2,523** systematically. Find the interval **k**, valid random starts, and the first four IDs if the random start is **17**.

INTERVAL, STARTS AND FIRST FOUR IDS

OPTIONAL

Equal chances for whom?

Slides 58-59

Six people: **A, B, C, D, E and F**. A computer selects one pair from **AB, CD or EF**. Each of these three pairs has an equal chance.

X1 What is each person's chance of selection?

X2 How many pairs could a simple random sample select?

X3 Is this process a simple random sample of two? Explain.
